

Hindi Etymology Culler Specifications

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1 Parameters

1.1 Memory Layout

The memory in the source data file for Hindi etymological origins will be formatted as follows if a 7C divider is adopted:

Hindi (Devanagari UTF-8 + intermixed ASCII)	Div.	Origin (ASCII)	EOL	Div.
E0 XX XX ... XX ... E0 XX XX	7C	XX ... XX	0D 0A	

Such a divider is rendered necessary by the fact that UTF-8 characters are intermixed with some of the Devanagari characters, resulting in the presence of ASCII characters (less than or equal to 0x7F) rather than Devanagari characters (greater than 0x7F) being unable to effectively differentiate the two fields, and Microsoft Access's inability to export data without dividers.

1.2 Culling Requirements

The following are necessary to cull the data in the set:

1. Etymology data must be calculated for words that will often recur in multiple consecutive rows.
 - This will necessitate storing the second and third bytes in a char[] and comparing them to the current string¹, ensuring the length of the two match to prevent erroneous substring matching. This will be more computationally efficient than shortening the data into one byte per character (though this is feasible).
2. ASCII characters occasionally appear in the Hindi field among the predominantly Devanagari UTF-8 characters.

¹The first byte of any given character may be neglected since it is always E0. These bytes may be filled once after allocating the buffer so that it can continually be copied into the destination file.

- Accordingly, while copying from the Hindi field, characters determined to not be Devanagari UTF-8 due to inequality with E0 must additionally be compared to 7C to establish if they demarcate field dividers, are ASCII characters within the Hindi field, or are irregularities (see [3]).
3. Malformed data is present within the source file; unfortunately, the extent to which and heterogeneity with which this occurs is unknown.
 - One known example is a line with both fields entirely blank; thus, it merely comprises a 7C divider.
 - The culling algorithm should thus be sufficiently resilient to appropriately discard anomalies such as empty fields; moreover, it may be advisable to validate that Devanagari characters are valid as well, such as by discarding rows if a byte that is unequal to E0 but exceeds 0x7F occurs three bytes after the initial E0 byte of the preceding Devanagari character or if a continuation byte of a Devanagari character is less than or equal to 0x7F, both of which would indicate malformed Devanagari.
 4. An association will be counted if the origin data *contains* one of the defined qualitative buckets² rather than merely equaling it.
 - For computational efficiency and algorithmic simplicity, an assumption that the target word is properly capitalized will be adopted.
 5. Words with etymological origins involving multiple defined qualitative buckets will be rejected from the final data.
 - The established etymological origin of a word will be encoded through a state integer, perhaps the first letter of each qualitative bucket as ASCII or another integer mapping as appropriate. This will allow discards on the basis of simple inequalities and facilitate formatting and storage in the final dataset.

²Sanskrit, Persian, or English